

CRM Account Data Cleaning – Study Guide

24 September 2026 · Sacha

The Dynamics 365 Account export (8,622 organisations, 339 columns) is structurally sound but has content problems: about 60% of the columns are empty or always hold the same value, and there are duplicates, test records and inconsistent formatting. The recommended approach is to profile and plan on the export, use AI to propose changes, have a person approve them, and apply them in Dynamics.

1. The dataset at a glance

`Account_data.csv` is a Microsoft Dynamics 365 CRM export of organisation ("Account") records. It is well-formed: every row has the right number of columns and no quotes are broken. All of its problems are in the content.

Measure	Value
Rows (organisations)	8,622
Columns	339
File size	about 21 MB
Records created	May 2022 to September 2026
Active / inactive	8,165 / 457
Already merged into another record	425
Columns completely empty	137
Columns with the same value on every row	about 76
Columns holding meaningful data	about 40–60
Blank cells stored as the text <code>NULL</code>	about 1.9 million

The largest sources of records are the Stripe integration (about 2,900), manual entry by staff, the CRM team and the website portal integration. Knowing where records come from matters, because an integration that writes bad data will keep writing it.

Useful columns to keep in mind: `accountid` (the unique record ID, needed to update records back in Dynamics), `name`, the `address1_*` fields, `websiteurl`, social media fields, `cuk_organisationtype*`, `cuk_localauthority*`, `statecode`, `merged/masterid`, `createdon`, `modifiedon` and `createdby`.

2. Data quality issues found

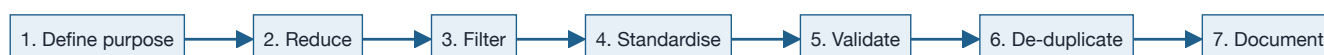
The problems fall into five themes. The biggest quick win is removing unused columns; the biggest risk is duplicates.

Theme	Issue	Count	Example
Structure	Empty columns	137	<code>address2_*</code> , <code>fax</code> , <code>creditlimit</code>

Theme	Issue	Count	Example
Structure	Columns with one value on every row	~76	credithold = No, currency = £
Structure	Code/label/"yomi" copies of one field	dozens	statecode 0/1 and statecodename Active/Inactive
Structure	Blanks stored as text NULL	~1.9m cells	A spreadsheet counts them as filled
Duplicates	Exact duplicate names	404 groups (1,120 rows)	Little Riders × 9
Duplicates	Duplicates after ignoring case, spacing, punctuation	558 groups (1,480 rows)	CyclingUK / Cycling UK ; C C Giro / CCGiro
Invalid records	Job titles stored as organisation names	~230	THE WARDEN × 185, all created by the Stripe integration on 22 June 2022
Invalid records	Test or dummy records	~125 names, 49 test cities	Test Org 2 , TEST Dixisset , city Test Town , "Lorem ipsum"
Formatting	Country written many ways	27 variants	UK 2,452; United Kingdom 885; Untied Kingdom ; F , D , CDN
Formatting	Missing country but has a UK postcode	1,843	
Formatting	Postcodes not in standard UK format	157	Cut short (SW1A 0A), foreign, test
Formatting	ALL CAPS or all lowercase names	492 / 73	SUSTRANS , phoenix group
Formatting	Leading or trailing spaces	381 names, hundreds of addresses	St Paul's Youth Forum
Formatting	Garbled characters (encoding)	17 names	Martinâ€™s Bicycle Services
Formatting	Postcode present but city missing	2,522	
Formatting	county and stateorprovince duplicate each other	95% equal where both are filled	
Validity	Non-website values in websiteurl	190 Marketing ; 93 social links or emails	Page titles such as HOME – Inspired Neighbourhoods
Validity	Wrong platform in social fields	a few	LinkedIn URL in the Instagram field
Validity	Impossible founding years	42	3000, 1500, 0, 6
Validity	Implausible numbers	a few	1,198,687 employees; 96,000 volunteers
Validity	Placeholder contact details	a few	none published , [Messenger] , phone 123455
Validity	Registration numbers in mixed formats	302 filled	SC 035350 , Sc047302 , SC040506

3. The general cleaning workflow

Data cleaning follows a fixed order. Reduce first so that every later step handles less data, and de-duplicate last because matching only works on standardised values.

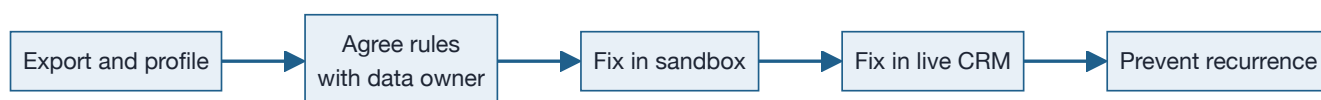


1. **Define the purpose.** Decide what question the clean data must answer. This decides which columns survive.
2. **Reduce.** Drop empty and single-value columns, keep one of each code/label pair, and convert `NULL` to real blanks.
3. **Filter.** Set aside test, merged and (possibly) inactive records. Flag them rather than deleting them.
4. **Standardise.** Trim spaces, fix letter case, map countries to one list, format postcodes and repair garbled characters.
5. **Validate.** Check URLs, emails, years and numbers against plausible ranges, and flag anything outside them.
6. **De-duplicate.** Match on normalised name plus postcode. A person reviews each match; nothing merges automatically.
7. **Document.** Keep a data dictionary (what each kept column means) and a change log (what changed, why, and by which rule).

Two principles apply throughout: **never overwrite the original file**, and **flag before you delete**.

4. Where cleaning happens: export vs CRM

Professionals analyse on an export and fix in the CRM. The CRM is the source of truth, so a clean copy on a laptop helps nobody else until the fixes reach Dynamics.



1. **Export and profile.** Take a snapshot and analyse it offline. It is fast and safe, because you cannot break anything.
2. **Agree the rules.** The data owner (usually the CRM team) signs off fixes, because merges and deletions affect fundraising, membership and marketing.
3. **Fix in the CRM, in a controlled way.** Test in a sandbox first, take a backup export, work in batches and log every change.
4. **Prevent recurrence.** Cleaning without prevention means doing it again next year.

Dynamics 365 tools for applying fixes

Tool	Best for	Notes
Bulk edit / Edit in Excel Online	A few hundred records	Change a view's values in Excel and save back
Export to Excel → edit → re-import	Large standardisation jobs	The hidden <code>accountid</code> makes the import update existing records rather than create new ones
Duplicate detection rules + Merge	Duplicates	Merge moves linked contacts, activities and donations onto the surviving record
Bulk delete jobs	Test records	Filter-based; check the filter first
XrmToolBox (e.g. Bulk Data Updater), Power Automate	Complex or repeated updates	Usually need CRM admin rights
Address validation services	Postcodes and addresses	Checks against the Royal Mail address file

Never de-duplicate in a spreadsheet and re-import. Deleting a record outside Dynamics' Merge orphans or loses everything linked to it.

Preventing recurrence

- Replace free-text fields with fixed lists, e.g. Country as a dropdown.
- Add field validation and required fields.
- Turn on duplicate detection rules so staff are warned before creating a near-duplicate.
- Fix the integrations. The Stripe and Portals integrations created thousands of records, including the 230 job-title "organisations". If an integration keeps writing bad data, it will overwrite fixes.

Two kinds of cleaning

Cleaning the source fixes Dynamics for everyone. **Cleaning for analysis** makes a tidy copy for one report (e.g. in Power BI or Power Query) and leaves the CRM alone. Always state which one you are doing.

In an apprenticeship you will probably not have write access to the live CRM, and that is fine: cleaning the export, documenting the rules and recommending CRM fixes is the correct professional first step.

5. The role of AI

AI can do most of the analysis and prepare the changes, but a person should approve them and a controlled process should apply them. The pattern is: **AI proposes** → **person approves** → **controlled process applies** → **AI verifies**.

Stage	How much AI can do	Notes
Profiling and finding problems	Almost all	Fast, thorough, repeatable
Drafting cleaning rules	Most	A person checks the rules make business sense
Judgement calls on messy values	AI's strongest area	Is C C Giro the same club as CCGiro ? Is Marketing a website? Which country is F ?
Producing the change set	Most	A file of record ID, field, old value, new value, reason: the key hand-off document
Applying changes in Dynamics	Possible, but gate it	See risk tiers below
Checking afterwards	Almost all	Re-export, re-profile, compare with the change set

Can AI change the CRM directly?

Technically, yes, increasingly. Copilot is built into Dynamics 365 and Power Platform. Dataverse (the database under Dynamics) has an API and a Microsoft-provided MCP server, which lets AI agents read and write records within the permissions of the account they use. Alternatively, AI writes a script that a person reviews and runs.

Risk tiers

Tier	Examples	Level of control
Green low risk, reversible	Trimming spaces, fixing case, standardising country, formatting postcodes	Automate heavily after reviewing a sample; log old values so they can be undone
Amber medium risk	Deactivating test records, correcting websites and social fields	AI proposes, a person skims the list, applied in batches

Tier	Examples	Level of control
Red high risk, hard to undo	Merging duplicates, deleting records	A person approves every group; a wrong merge combines two real organisations' history

Why a person stays involved

- **AI makes confident mistakes.** "Worcester St Johns CC" and "Worcester St John's Cycling Club" might be one club or two. Local knowledge decides.
- **Data protection.** The AI tool sees personal data, so it must be an organisation-approved tool with the right data agreement.
- **Accountability.** Someone owns each decision, and there is a record of what changed and why.
- **Permissions.** An AI agent should use a restricted account, tested on a sandbox first, never a full admin account.

6. Recommendation for this dataset

Treat this as a **cleaning-for-analysis project that ends in recommendations for the CRM team**. Work on a copy of the export, use AI to write rules and scripts rather than to read the full file, and hand over a reviewed change set.

Ground rules

- **Keep the original file untouched.** Save each stage as a new file (`01_reduced.csv` , `02_filtered.csv` ...).
- **Do not paste the full file into an AI chat.** It holds real emails, phone numbers, addresses and staff names, and at 21 MB it is too large anyway. Give the AI the column profile plus a 20–50 row sample with names, emails and phone numbers replaced by placeholders. Check Cycling UK's AI and data protection policy first.
- **Have the AI write the transformation** (a pandas script or Power Query steps) and run it yourself on the full file locally.
- **Add flag columns rather than deleting rows**, e.g. `flag_test` , `flag_job_title_name` , `flag_possible_duplicate` , `flag_invalid_postcode` .
- **Keep `accountid` on every row** so any fix can be traced back to, and applied in, Dynamics.

Phased plan

Phase	What to do on this file	Output
1. Purpose	Agree 2–3 questions with your manager, e.g. "How many active clubs and groups are there per region?"	One-paragraph brief
2. Reduce	Convert NULL to blank; drop the 137 empty and ~76 constant columns; keep only the <code>*name</code> label of each code/label pair; drop <code>yominame</code> and <code>_base</code> copies	Working set of ~40–60 columns + data dictionary
3. Filter (flag)	Flag ~125 test names and test addresses; flag ~230 job-title names (<code>THE WARDEN</code> etc.); flag 425 merged and 457 inactive records	Flag columns; counts per flag
4. Standardise (green)	Trim spaces; title-case ALL CAPS and lowercase names (with exceptions for acronyms like CIC, CC, NHS); repair the 17 garbled names; map 27 country spellings to one list; fill country where the postcode is UK; uppercase and space postcodes; merge <code>county</code> and <code>stateorprovince</code>	Cleaned columns + change log

Phase	What to do on this file	Output
5. Validate (amber)	Blank or flag the 190 Marketing websites; move social links out of websiteurl; check each social field matches its platform; flag the 42 impossible years and extreme employee/volunteer counts; standardise registration numbers	Validation report
6. De-duplicate (red)	Build a match key of normalised name + postcode; have AI score likely pairs; review every group by hand	Duplicate review sheet for the CRM team
7. Hand over	Summarise findings, the change set, and prevention ideas (country dropdown, duplicate rules, fixing the Stripe import)	Report + meeting with the CRM team

Tooling

- **Python with pandas** is recommended if your course allows it: scripts are repeatable on next month's export and act as a record of every change. Install it in a virtual environment inside a project folder.
- **Excel with Power Query** is a good alternative if your course is Excel-based. Power Query also records each step.

Questions for the CRM team

- ☐ Why were the THE WARDEN records created by the Stripe integration in June 2022, and are they real organisations?
- ☐ Are test records safe to deactivate, or are some used for training?
- ☐ Which fields are genuinely in use, and which are left over from old projects?
- ☐ Is there a sandbox environment, and are duplicate detection rules switched on?
- ☐ Which AI tools are approved for use with CRM data?

7. Building a reusable prompt template

A good data-cleaning prompt has seven parts. Keep the first three the same across every prompt, and change the task, rules and output for each cleaning stage.

Part	Purpose	Example for this dataset
1. Role	Sets the expertise and tone	"You are a data quality analyst experienced with Microsoft Dynamics 365 CRM data."
2. Context	Why the work matters and who uses it	"Cycling UK's Account table of partner organisations, used for the Community Cycle Club and events programmes."
3. Input	Exactly what the AI is given	Column profile + anonymised 30-row sample, as CSV
4. Task	One clear job per prompt	"Propose standardisation rules for the address columns."
5. Rules and constraints	What it must and must not do	"Flag, never delete. UK postcodes use the format AA9A 9AA . Keep accountid unchanged."
6. Output format	Makes the answer usable and checkable	"A table with columns: rule, column, before, after, rows affected, risk (green/amber/red)."
7. Check and assumptions	Surfaces uncertainty	"List any assumptions and anything you are unsure about. Do not guess."

Good practice

- **One stage per prompt.** A prompt that asks for everything at once gets shallow answers.
- **Show, don't describe.** A small sample of real (anonymised) rows beats a paragraph describing them.
- **Give examples of the answer you want**, e.g. one completed row of the output table.
- **Use placeholders** like `{{DATASET_NAME}}` so the same template works for the Contacts table next.
- **Ask for code, not cleaned data**, when the file is large or sensitive.
- **Chain the prompts.** Paste the approved output of one stage into the next as input.
- **Save every prompt and response** as evidence of your method.

8. Example prompt templates

Start with the master template, then use one stage prompt at a time. Text in `{{double braces}}` is a placeholder to fill in.

Master template

```
## Role
You are a data quality analyst experienced with Microsoft Dynamics 365 CRM exports.

## Context
Dataset: {{DATASET_NAME}} – {{ONE_LINE_DESCRIPTION}}.
Purpose of the clean data: {{BUSINESS_QUESTION}}.
The cleaned data will be used by: {{AUDIENCE}}.

## Input
{{COLUMN_PROFILE_OR_SAMPLE}}
(Personal data has been replaced with placeholders.)

## Task
{{ONE_SPECIFIC_TASK}}

## Rules
– Never delete records; add a flag column instead.
– Never change the record ID column ({{ID_COLUMN}}).
– {{EXTRA_RULES}}

## Output format
{{OUTPUT_FORMAT}}

## Check
List every assumption you made and every value you were unsure about.
If information is missing, ask rather than guess.
```

Prompt A: Reduce columns and write a data dictionary

```
## Task
Using the column profile below (column name, % filled, number of distinct values, sample values),
sort every column into KEEP, DROP or REVIEW.
- DROP: empty, single-valued, or a duplicate of another column (e.g. a code with a matching label column).
- KEEP: useful for {{BUSINESS_QUESTION}}.
- REVIEW: unclear.
For each KEEP column, write a one-line plain-English description.

## Output format
A table: column | decision | reason | description (KEEP only).
Then a count of columns in each group.
```

Prompt B: Standardise a messy field (countries)

```
## Task
Below is every distinct value of address1_country with its row count.
Propose a mapping to a single standard list of country names.

## Rules
- Use full English names (e.g. "United Kingdom", "France").
- England, Wales, Scotland and Northern Ireland map to "United Kingdom";
  also propose a separate "nation" column that keeps them.
- Single letters are international vehicle codes (F = France, D = Germany).
- Mark any mapping you are not confident about as REVIEW.

## Input
UK 2452 | United Kingdom 885 | England 55 | Wales 29 | F 7 | USA 7 | AUS 5 | B 5 |
NL 4 | DK 4 | I 3 | Northern Ireland 2 | SW 2 | IRL 2 | PL 2 | CH 2 | RS 2 | D 2 |
Untied Kingdom 2 | United Kindgrom 1 | GBG 1 | NZ 1 | CDN 1 | Malta 1 | Ireland 1 |
United State 1 | US 1

## Output format
A table: original value | standard value | nation (if UK) | confidence (high/REVIEW) | rows
affected.
```

Prompt C: Write the cleaning script

```
## Task
Write a Python script using pandas that applies the approved rules below to {{FILE_NAME}}.

## Approved rules
{{PASTE_APPROVED_RULES_FROM_PROMPTS_A_AND_B}}

## Rules for the script
- Read the CSV with na_values="NULL" and encoding="utf-8-sig".
- Never overwrite the input file; write to {{OUTPUT_FILE}}.
- Add flag columns instead of deleting rows.
- Write a change log CSV: accountid, column, old value, new value, rule name.
- Print a summary of how many rows each rule changed.
- Comment each step in plain English for a beginner.

## Output format
The complete script, then a short explanation of how to run it.
```

Prompt D: Review possible duplicates

Task

Below are groups of Account records that share a normalised name or postcode.
For each group, decide whether they are the SAME organisation, DIFFERENT organisations, or UNSURE, and if SAME, suggest which record to keep.

Rules

- Prefer the record that is Active, most complete, and most recently modified.
- Branches of a national body (e.g. local groups of one charity) are DIFFERENT.
- If evidence is thin, answer UNSURE. Do not guess.

Input

```
{{GROUPS: accountid, name, postcode, city, website, createdon, modifiedon, status}}
```

Output format

A table: group | decision | record to keep | reason (one line) | confidence.

Prompt E: Verify the result

Task

Compare the column profile before and after cleaning (both below).
Confirm each approved rule had the expected effect, and list anything unexpected:
rows lost, new blanks, values changed that no rule covers.

Output format

A table: rule | expected effect | observed effect | pass/fail.
Then a list of issues still remaining, ranked by business impact.

9. Glossary

Term	Meaning
Account	Dynamics 365's name for an organisation record
accountid (GUID)	The unique ID of each record; needed to update records back in Dynamics
Code / label pair	Dynamics stores a choice both as a number (statecode = 0) and as text (statecodename = Active)
Yomi name	A phonetic spelling field for Japanese names; always a copy here and safe to drop
Dataverse	The database underneath Dynamics 365 and the Power Platform
Sandbox	A test copy of the CRM where changes can be tried safely
Data profiling	Measuring each column: how full it is, how many distinct values, typical values, outliers
Data dictionary	A list of kept columns with a plain-English description of each
Change log	A record of every change: record, column, old value, new value, rule
Normalisation (of text)	Removing case, spaces and punctuation so near-identical values match
Mojibake	Garbled characters (â€™ instead of ') caused by reading text in the wrong encoding
pandas	A Python library for working with tables of data in code
Virtual environment	An isolated folder for Python packages that does not affect the rest of the computer

Term	Meaning
Power Query	Excel and Power BI's built-in tool for recorded, repeatable data cleaning
MCP (Model Context Protocol)	A standard that lets AI agents connect to systems such as a CRM
Human in the loop	A person reviews and approves AI output before it takes effect